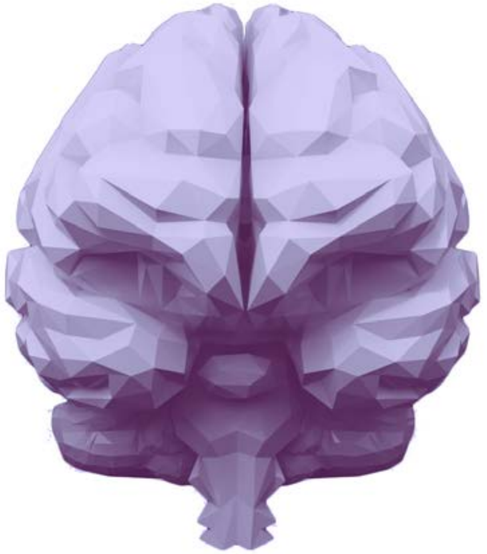
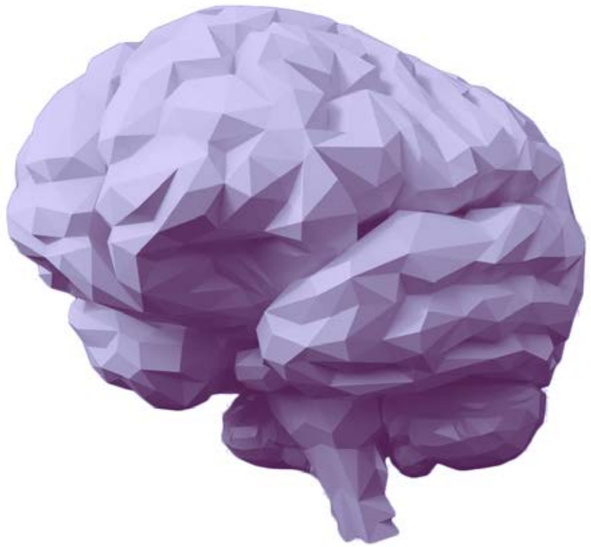


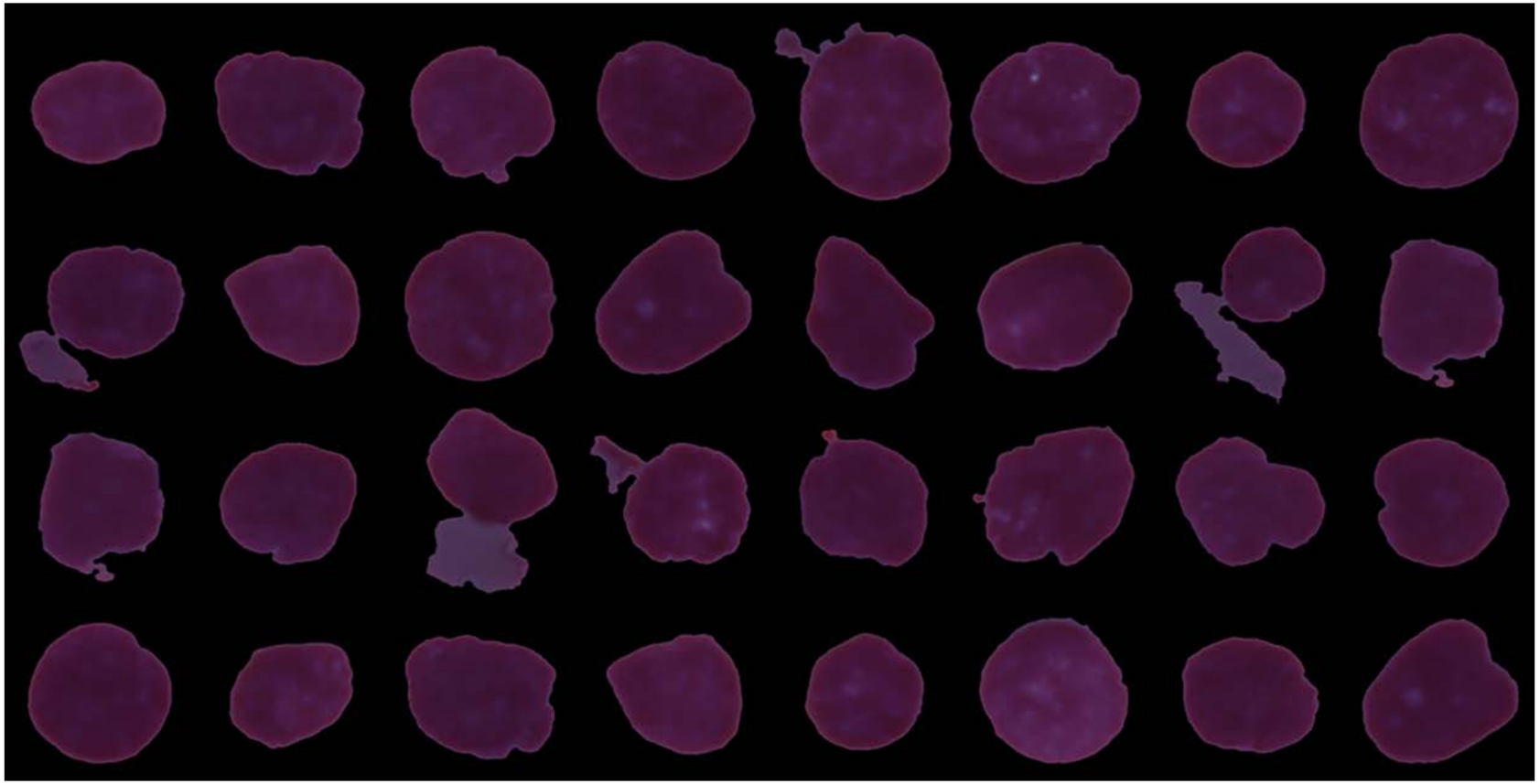




# ANNa

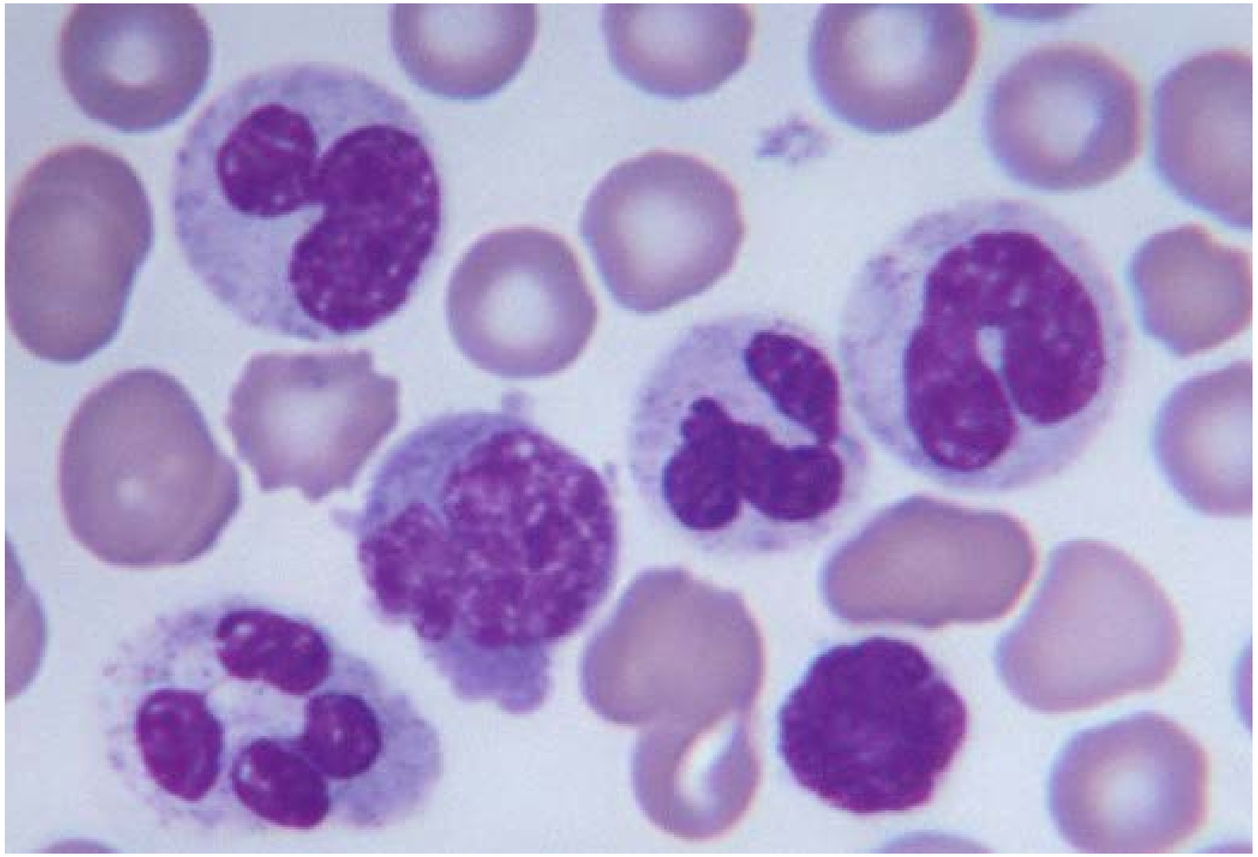
Design: Ruggero Agnolutto

# GOAL



The **goal** is to create a model able to **distinguish Leukemia from Healthy cells**.

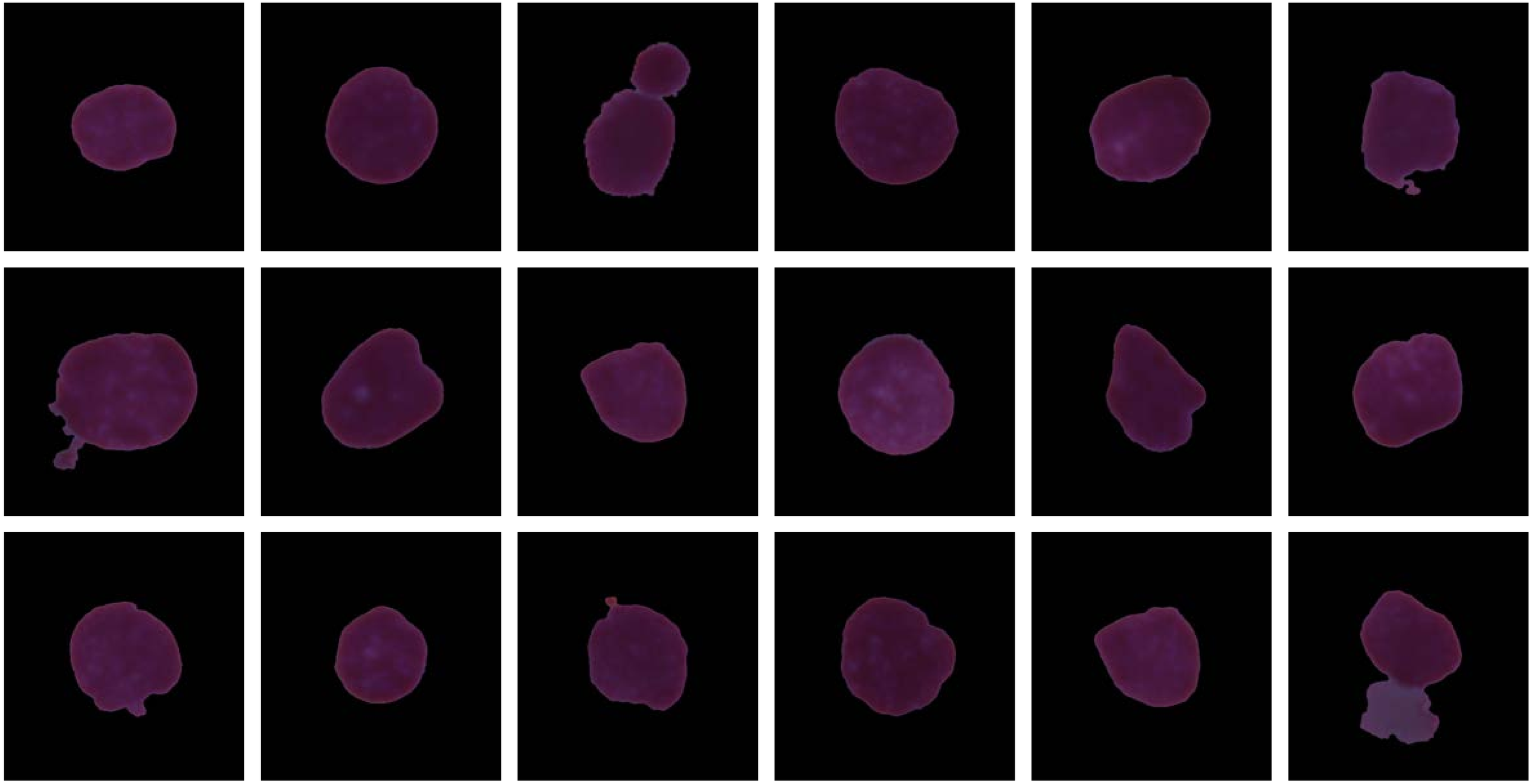
# LEUKEMIA



Leukemia is a malignant tumor of blood cells, specifically the white ones (leukocytes).

Its more common form, the **Acute lymphoblastic leukemia** (ALL) represents 25% of all pediatric cancers.

# MORPHOLOGICAL SIMILARITIES



# MORPHOLOGICAL SIMILARITIES









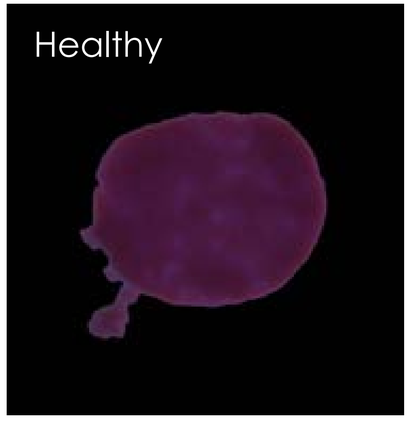


















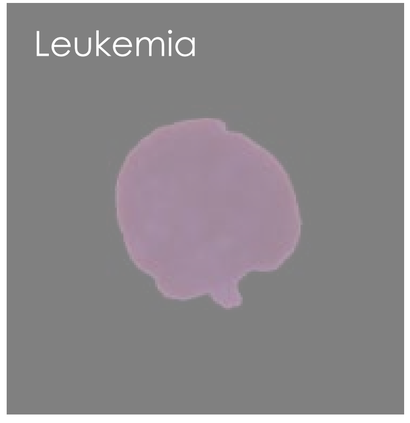



























# DATA STRUCTURE

## DATASET

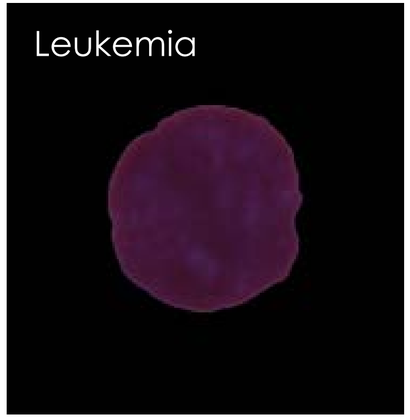
Images are obtained from a Kaggle dataset with more than 15000 images of **leukocytes**.

all = Leukemia
hem = Healthy cells







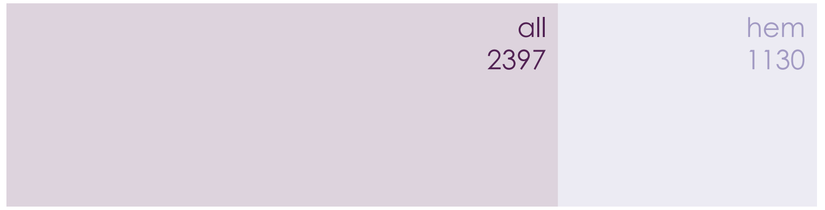




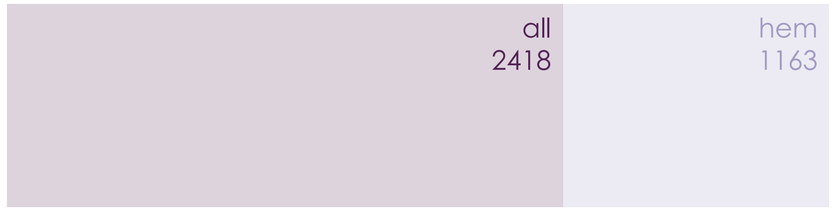


## TRAIN DATA / 10661 images

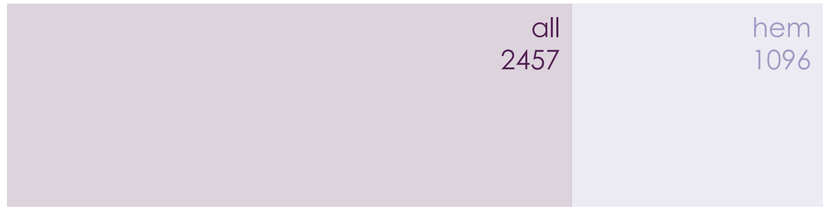
Fold 0




Fold 1




Fold 2




## VALIDATION DATA / 1867 images + labels




labels 0-hem 1-all

## % LEUKEMIA vs HEALTHY CELLS

Training and validation sets have almost the same percentage of images.

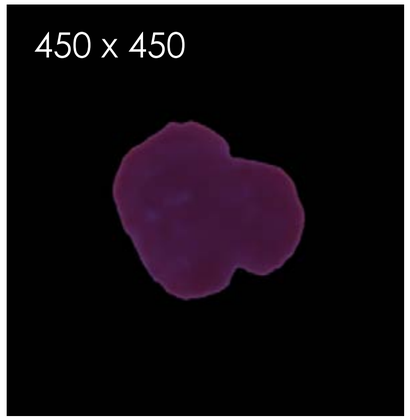
Training Set



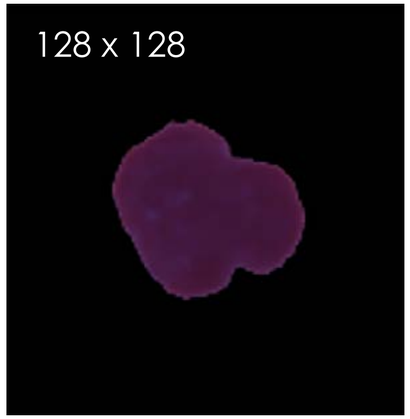

Validation Set

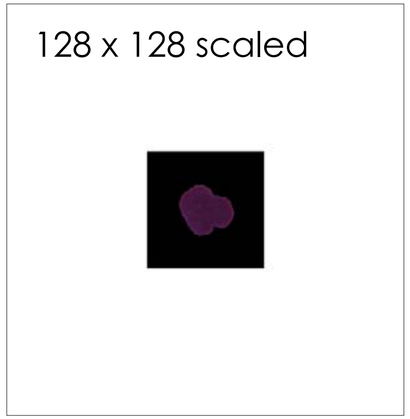


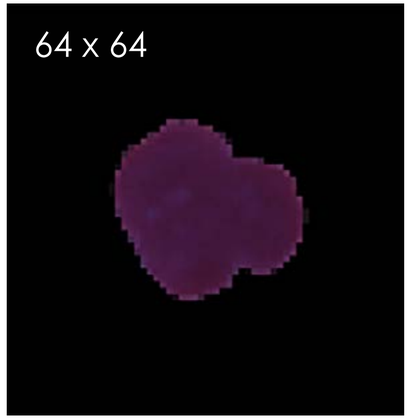
# IMAGE PREPROCESSING

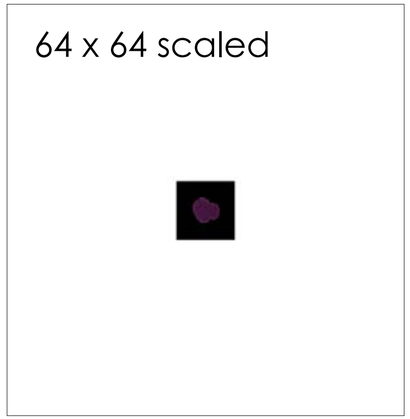



To reduce the computation time images have been scaled to a 128 x 128 resolution



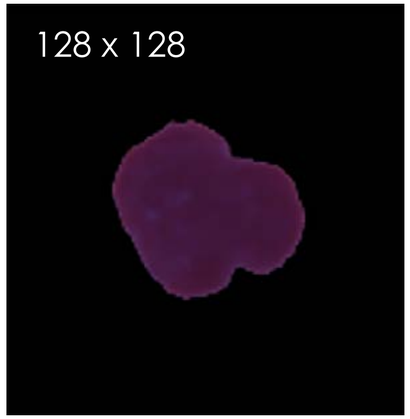



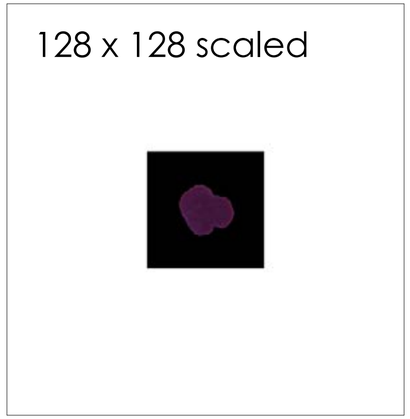



# IMAGE PREPROCESSING



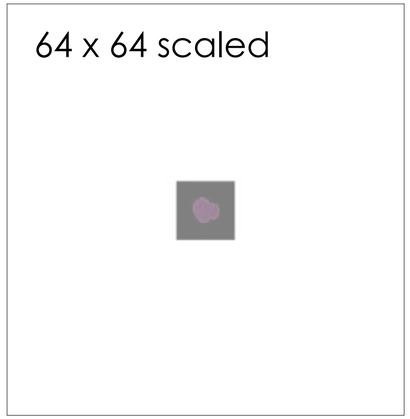

To reduce the computation time images have been scaled to a 128 x 128 resolution









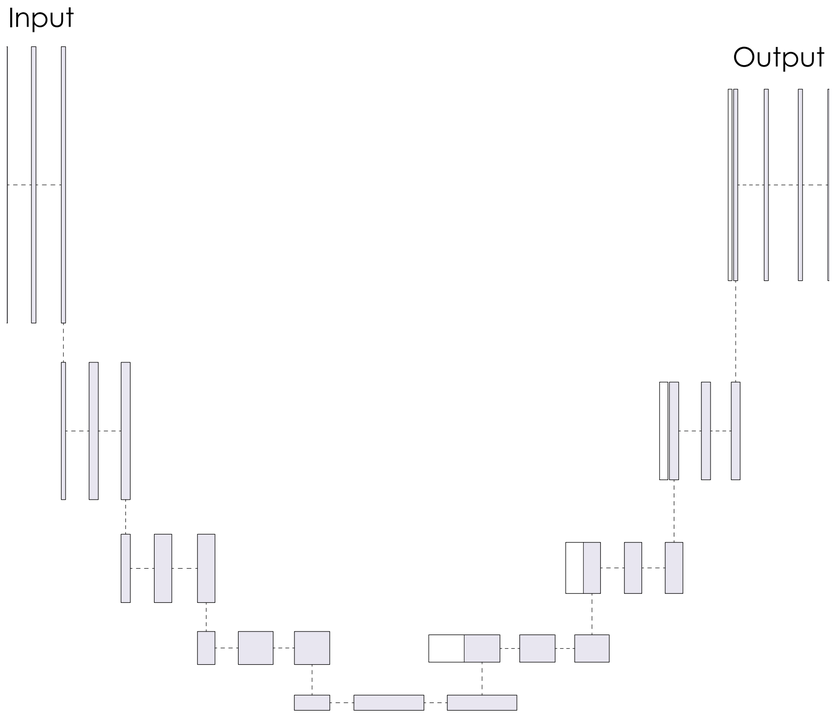
To achieve our goal, we create **ANNa**,
an Artificial Neural Network anti-Leukemia.



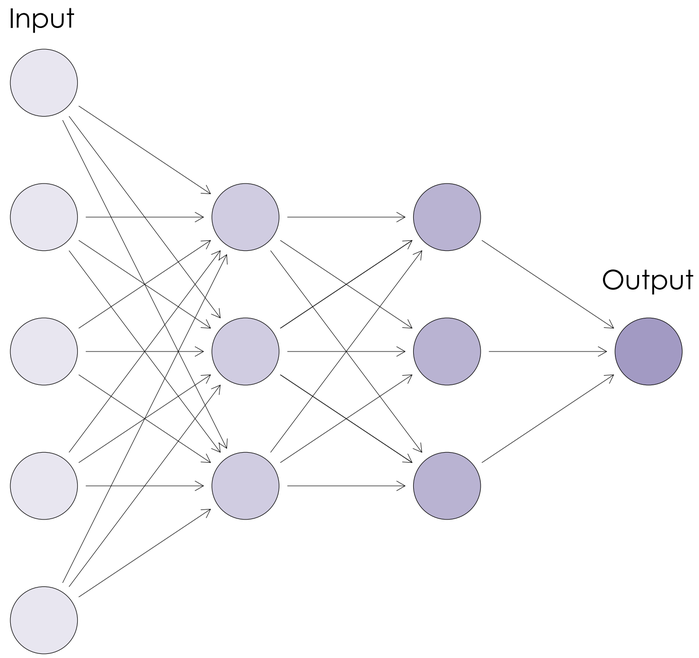
**Neural Networks** are computer systems modelled on the **human brain** and **nervous system**.

# ANNa's BRAIN

Made with: **Google Colab**






## U-Net

Convolutional Network for Biomedical Image Segmentation

Detection of boundaries

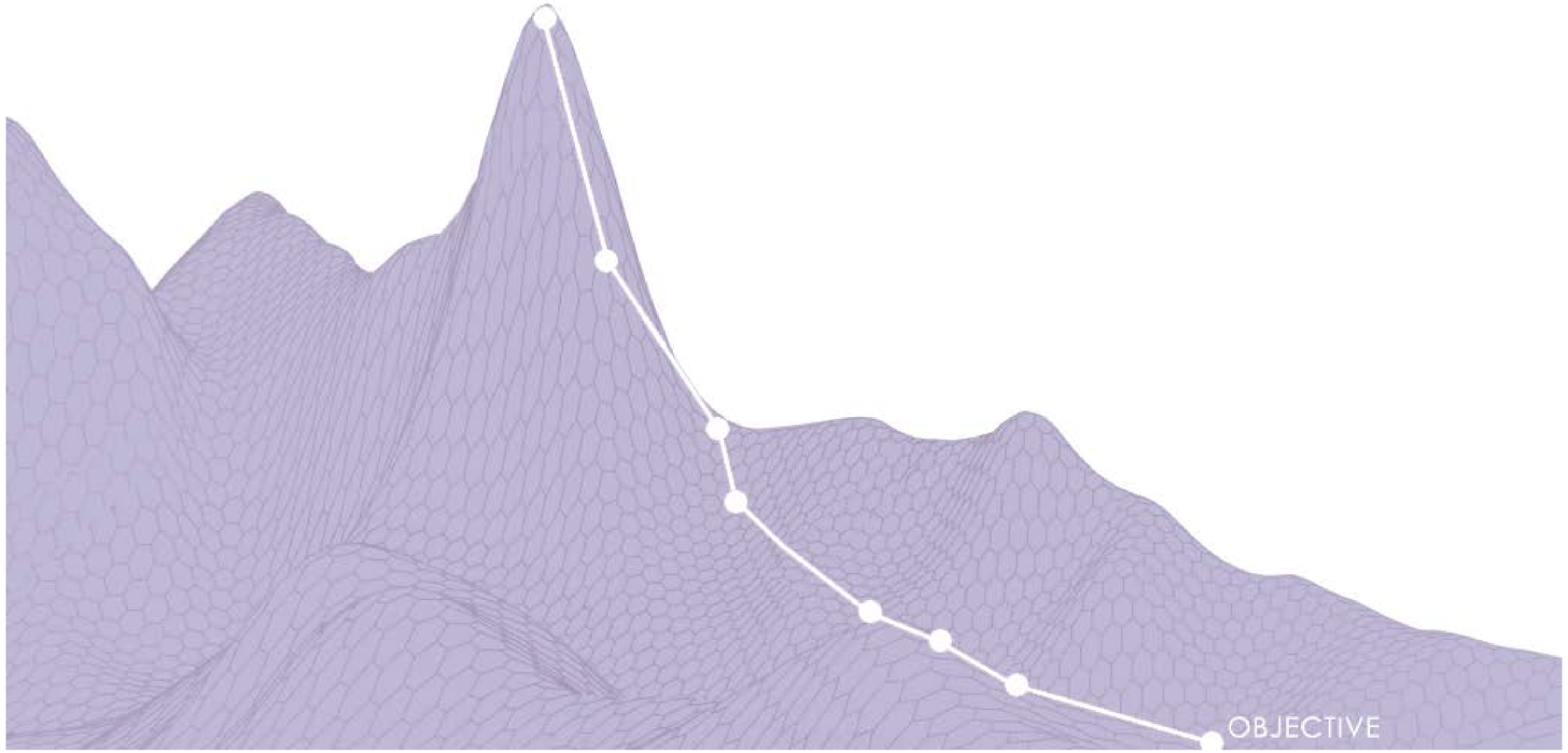
## Neural network

Binary classification

Healthy / Leukemia cells

# OPTIMIZATION

The aim of deep learning machines like ANNa is to create a model that can **predict** real data.
**Better the prediction, lowest the error** between predicted and actual data.

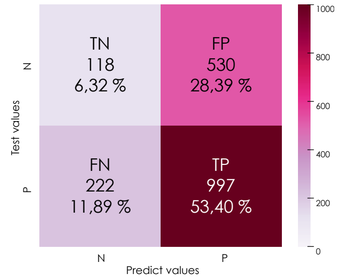


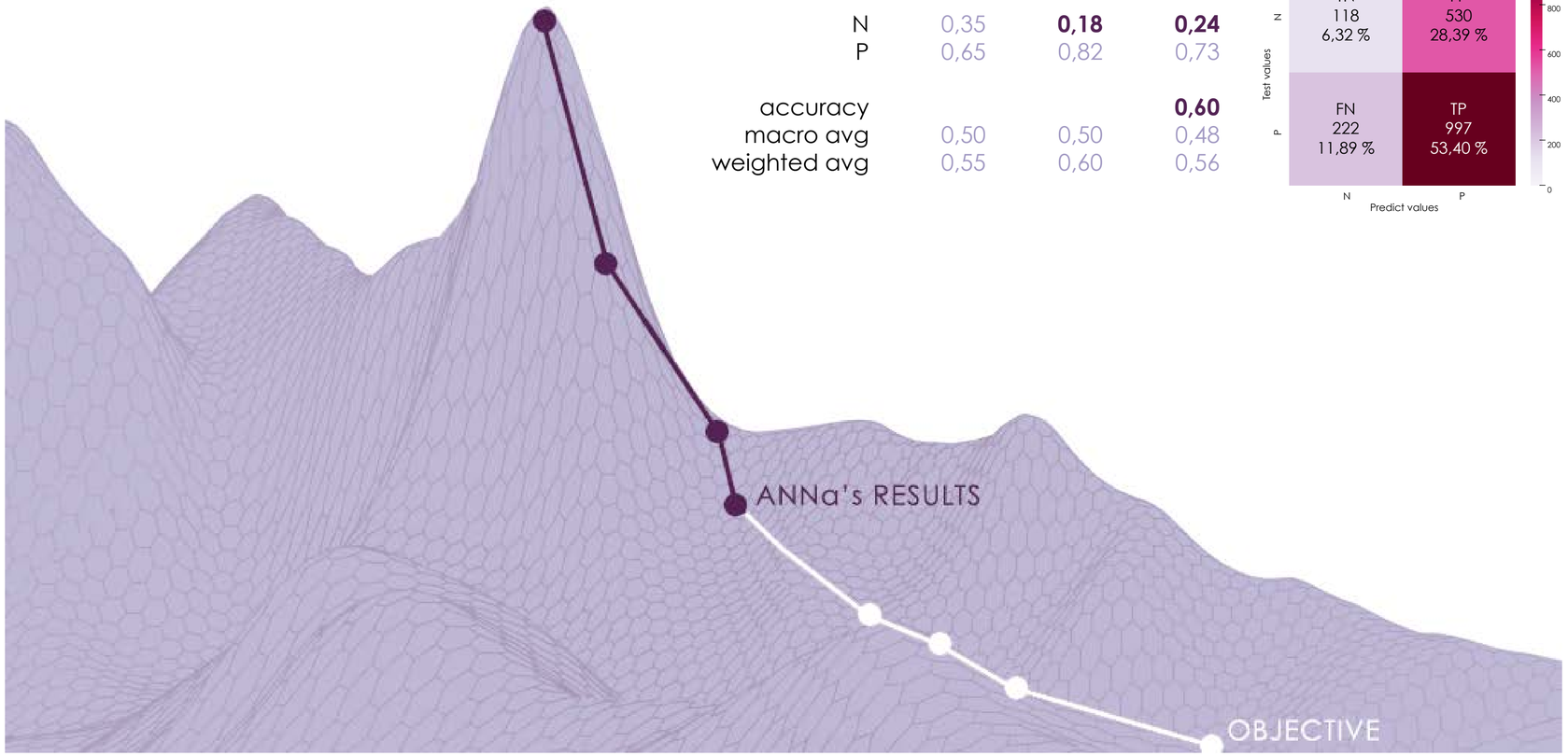

The **goal** is to to reach the lowest point that represents the **mininum error**.

# RESULTS

**ANNa** is able to recognize Diseased cells and is not able to clearly detect Healthy ones.

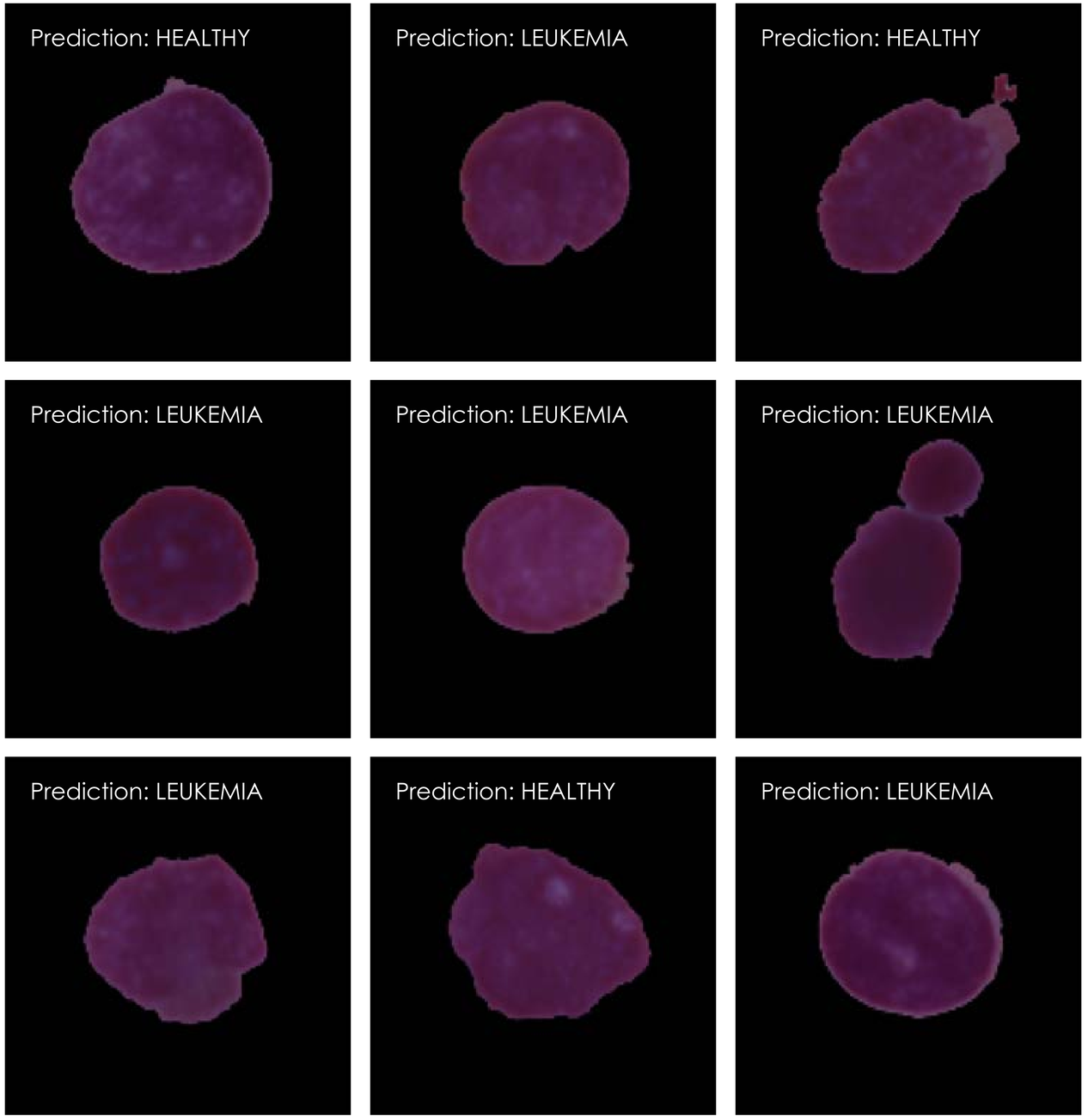
| | precision | recall | f1-score |
|---|---|---|---|
| N | 0,35 | **0,18** | **0,24** |
| P | 0,65 | 0,82 | 0,73 |
| accuracy | | | **0,60** |
| macro avg | 0,50 | 0,50 | 0,48 |
| weighted avg | 0,55 | 0,60 | 0,56 |



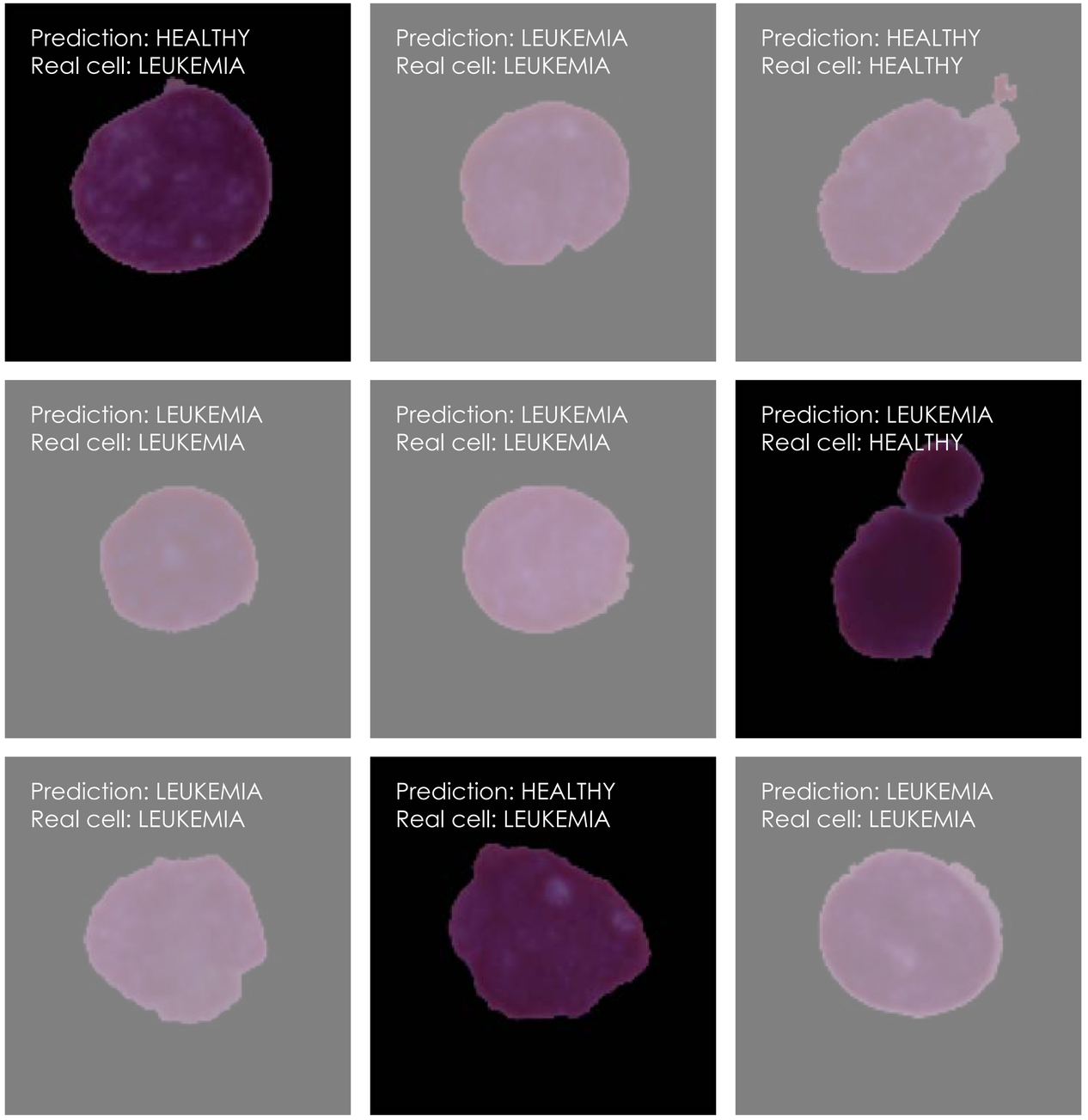




This means that it create several **false positives and negatives** that are very dangerous in medicine.

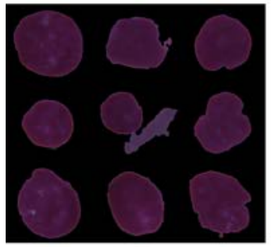
# RESULTS


Prediction: HEALTHY
Prediction: LEUKEMIA
Prediction: HEALTHY
Prediction: LEUKEMIA
Prediction: LEUKEMIA
Prediction: LEUKEMIA
Prediction: LEUKEMIA
Prediction: HEALTHY
Prediction: LEUKEMIA

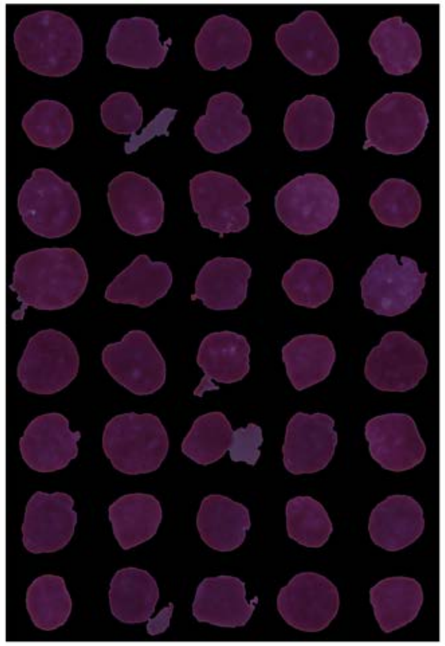
# RESULTS


Prediction: HEALTHY
Real cell: LEUKEMIA
Prediction: LEUKEMIA
Real cell: LEUKEMIA
Prediction: HEALTHY
Real cell: HEALTHY
Prediction: LEUKEMIA
Real cell: LEUKEMIA
Prediction: LEUKEMIA
Real cell: LEUKEMIA
Prediction: LEUKEMIA
Real cell: HEALTHY
Prediction: LEUKEMIA
Real cell: LEUKEMIA
Prediction: HEALTHY
Real cell: LEUKEMIA
Prediction: LEUKEMIA
Real cell: LEUKEMIA

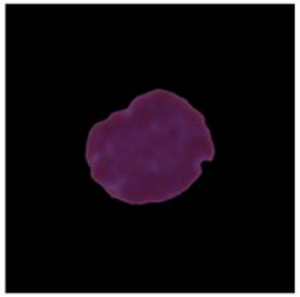
# FUTURE IMPROVEMENTS



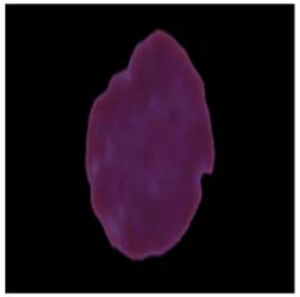
Actual dataset



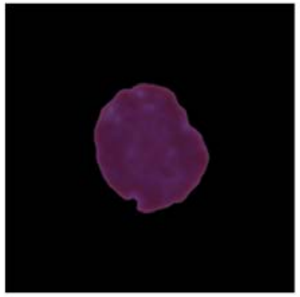
Ideal dataset



Original

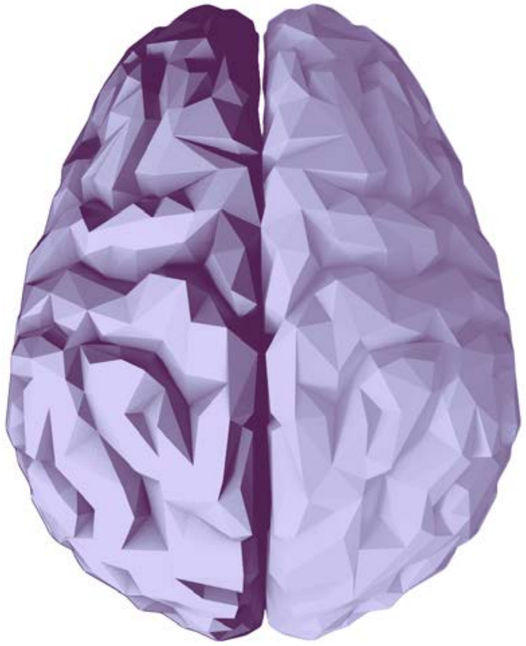


Stretched



Rotated

**U-Net**
Parameter optimization



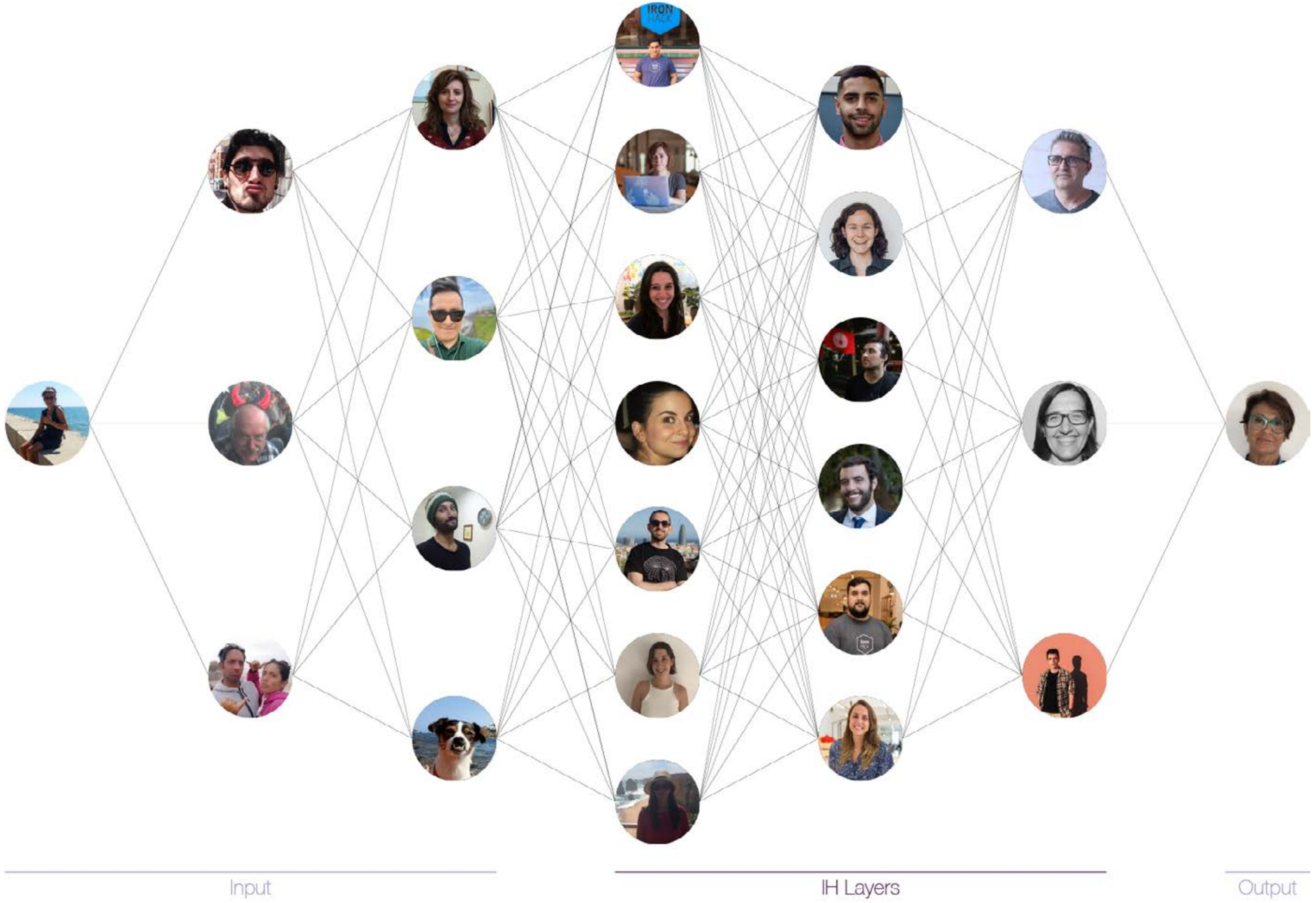
**Neural Network**
New layer structure

## BIGGER TRAINING SET

## DATA AUGMENTATION

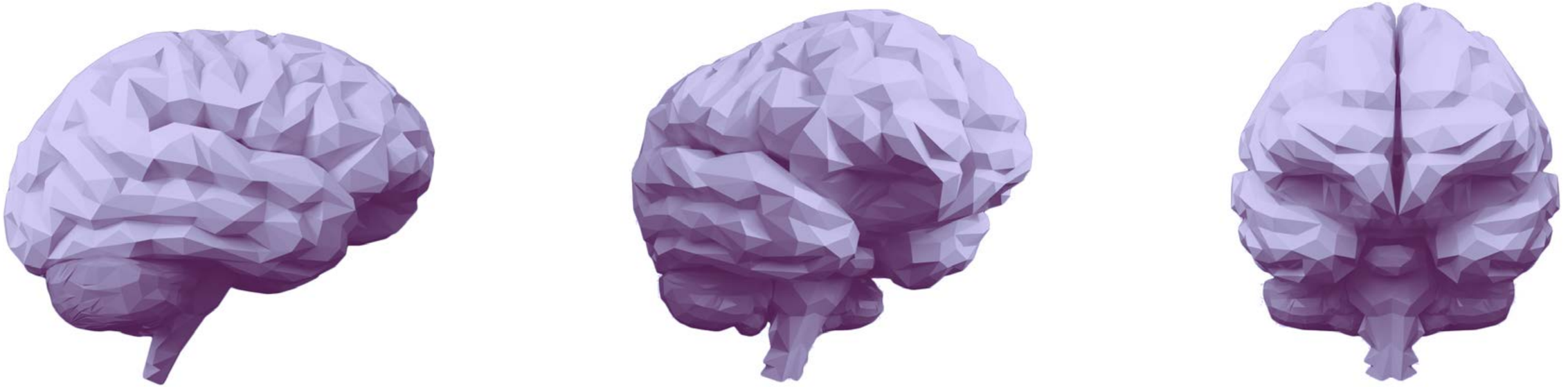
## NEW OPTIMIZATION

# THE CREW



# Thanks!

Veronica Agnolutto ANNa



*It's not some kind of miracle, cancer doesn't grow from yesterday to today. It's a pretty long process.*
*There are signs in the tissue, but the human eye has limited ability to detect what may be very small patterns.*

**Regina Barzilay**
first winner of the Squirrel AI Award for AI for the Benefit of Humanity